

Lab 5: Data Confidentiality in Information Security

1. Objective

To understand and implement the concept of confidentiality by using symmetric encryption to protect sensitive information such as passwords or data.

2. Tools & Requirements

Python 3

Google Colab

cryptography library (install using: pip install cryptography)

Any text editor or IDE (VSCode, PyCharm, Jupyter Notebook)

3. Setup Environment

1. Open your Python / Colab Terminal.

2. Install the required library:

```
pip install cryptography
```

4a. Single Message Encryption

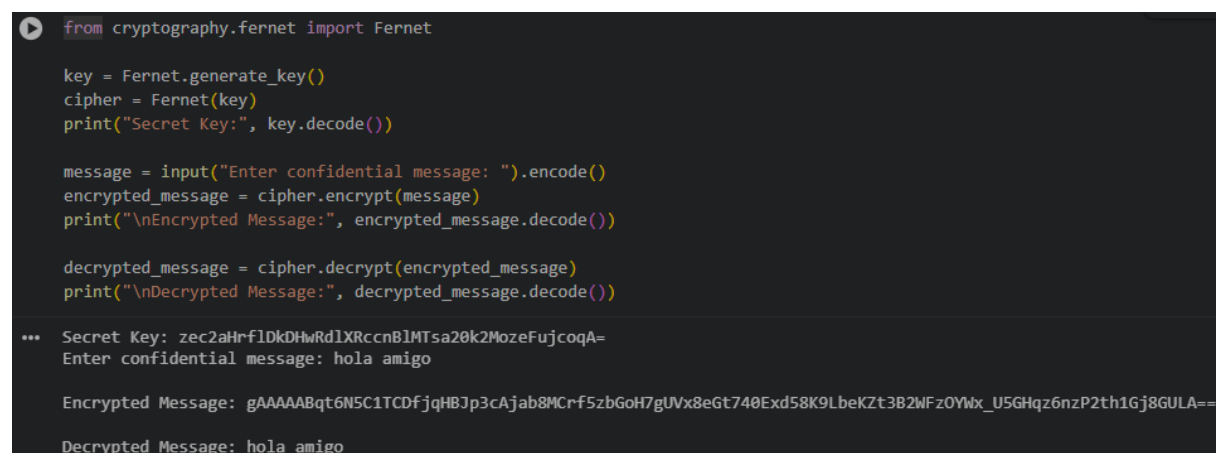
```
from cryptography.fernet import Fernet

key = Fernet.generate_key()
cipher = Fernet(key)
print("Secret Key:", key.decode())

message = input("Enter confidential message: ").encode()
encrypted_message = cipher.encrypt(message)
print("\nEncrypted Message:", encrypted_message)

decrypted_message = cipher.decrypt(encrypted_message)
print("\nDecrypted Message:", decrypted_message.decode())
```

Screenshot 1 4a: Single Message Encryption



```
from cryptography.fernet import Fernet

key = Fernet.generate_key()
cipher = Fernet(key)
print("Secret Key:", key.decode())

message = input("Enter confidential message: ").encode()
encrypted_message = cipher.encrypt(message)
print("\nEncrypted Message:", encrypted_message.decode())

decrypted_message = cipher.decrypt(encrypted_message)
print("\nDecrypted Message:", decrypted_message.decode())

*** Secret Key: zec2aHrflDkDHwRd1XRccnB1MTsa20k2MozeFujcoqA=
Enter confidential message: hola amigo

Encrypted Message: gAAAAABqt6N5C1TCDFjqHB3p3cAjab8Mcrf5zbGoH7gUVx8eGt740Exd58K9LbeKZt3B2WFz0YwX_U5GHqz6nzP2th1Gj8GULA==

Decrypted Message: hola amigo
```

4b. Batch Messages Encryption

```
from cryptography.fernet import Fernet

key = Fernet.generate_key()
cipher = Fernet(key)

messages = [
    "Patient ID 101: Diabetes report confidential",
    "Patient ID 102: COVID-19 test result confidential",
    "Patient ID 103: Prescription data confidential"
]

encrypted_messages = [cipher.encrypt(m.encode()) for m in messages]

print("\nEncrypted Batch:")
for e in encrypted_messages: print(e.decode())

print("\nDecrypted Batch:")
for e in encrypted_messages: print(cipher.decrypt(e).decode())
```

Screenshot 2 4b: Batch Messages Encryption



```
from cryptography.fernet import Fernet

key = Fernet.generate_key()
cipher = Fernet(key)

messages = [
    "Ths is info sec",
    "Abc XYZ",
    "Patient ID 103: Prescription data confidential"
]

encrypted_messages = [cipher.encrypt(m.encode()) for m in messages]

print("\nEncrypted Batch:")
for e in encrypted_messages:
    print(e.decode())

print("\nDecrypted Batch:")
for e in encrypted_messages:
    print(cipher.decrypt(e).decode())

***
Encrypted Batch:
gAAAAABqt6Fyisn0V7JX8yD7PkW6aqEyaviQteb0lVoDXw9AemwpM3Ac1LzSCDrZ4sgJ6ndoDlVljwEaATJTr7EeU31mm5YPXA==
gAAAAABqt6FyUGvyhvNefq7GpVzMUXv93VADdZvsc4J-vbMvgHdUz19SdlbGaBfuYQarGEQreABiEzUjhL1ajDZUrI7CELLR0g==
gAAAAABqt6Fynz5k8swcEkks65ldUe1cQyLcfSonjuD4vGdhnZka4YCbLonfv3g9Lgyt6GKKbsKNovcs1dR5QI8GKlVlE5jFb4dBwr4BiBj1UmGXBetzzidxLINMx8DhKm

Decrypted Batch:
Ths is info sec
Abc XYZ
Patient ID 103: Prescription data confidential
```

4c. Key Prompt Decryption

```
from cryptography.fernet import Fernet

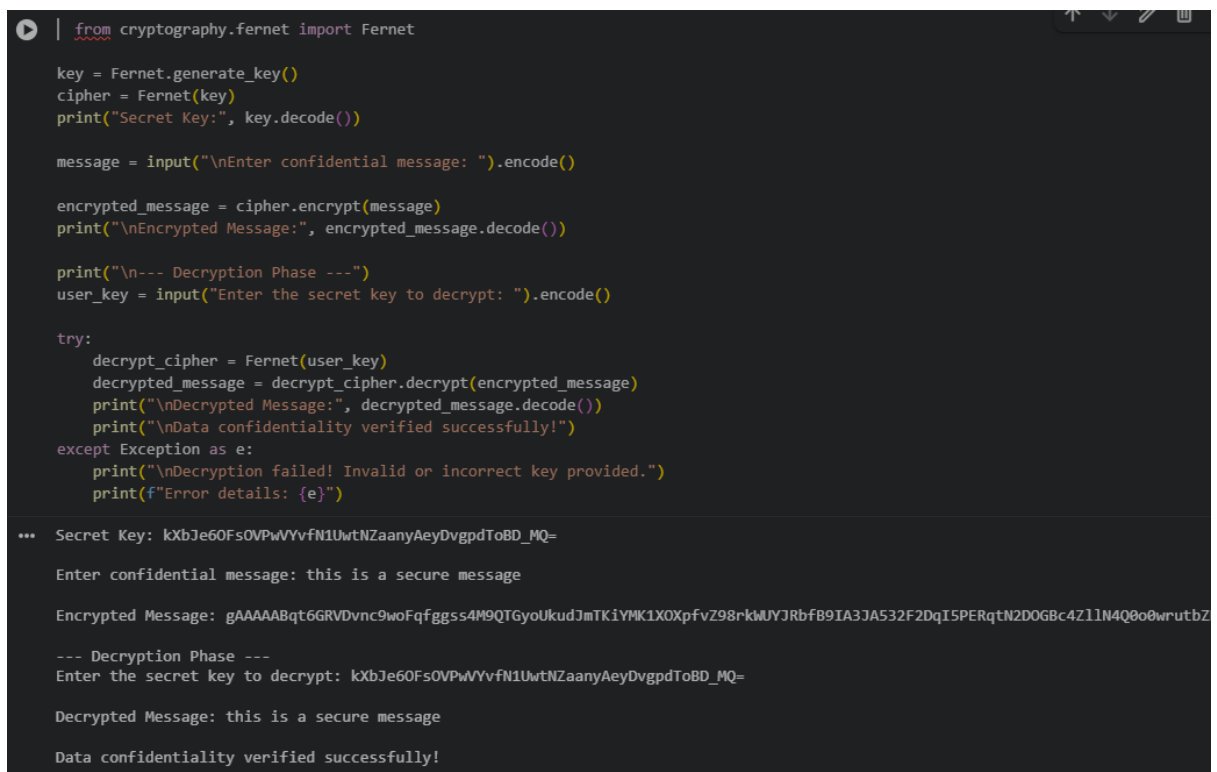
key = Fernet.generate_key()
cipher = Fernet(key)
print("Secret Key:", key.decode())

message = input("\nEnter confidential message: ").encode()
encrypted_message = cipher.encrypt(message)
print("\nEncrypted Message:", encrypted_message.decode())
```

```
print("\n--- Decryption Phase ---")
user_key = input("Enter the secret key to decrypt: ").encode()

try:
    decrypt_cipher = Fernet(user_key)
    decrypted_message = decrypt_cipher.decrypt(encrypted_message)
    print("\nDecrypted Message:", decrypted_message.decode())
    print("\nData confidentiality verified successfully!")
except Exception as e:
    print("\nDecryption failed! Invalid or incorrect key provided.")
    print(f"Error details: {e}")
```

Screenshot 3 — 4c: Key Prompt Decryption



```
from cryptography.fernet import Fernet

key = Fernet.generate_key()
cipher = Fernet(key)
print("Secret Key:", key.decode())

message = input("\nEnter confidential message: ").encode()

encrypted_message = cipher.encrypt(message)
print("\nEncrypted Message:", encrypted_message.decode())

print("\n--- Decryption Phase ---")
user_key = input("Enter the secret key to decrypt: ").encode()

try:
    decrypt_cipher = Fernet(user_key)
    decrypted_message = decrypt_cipher.decrypt(encrypted_message)
    print("\nDecrypted Message:", decrypted_message.decode())
    print("\nData confidentiality verified successfully!")
except Exception as e:
    print("\nDecryption failed! Invalid or incorrect key provided.")
    print(f"Error details: {e}")

... Secret Key: kXbJe60Fs0VPwVYvFN1UwtNZaanyAeyDvgpdToBD_MQ=

Enter confidential message: this is a secure message

Encrypted Message: gAAAAABqt6GRVDvnc9woFqfggss4M9QTGyoUkudJmTKiYMK1X0XpfvZ98rkWUYJRbfB9IA3JA532F2DqI5PERqtN2D0Gbc4Z1lN4Q0o0wrutbZ

--- Decryption Phase ---
Enter the secret key to decrypt: kXbJe60Fs0VPwVYvFN1UwtNZaanyAeyDvgpdToBD_MQ=

Decrypted Message: this is a secure message

Data confidentiality verified successfully!
```

Lab Task

Practice the code and attach screenshots above.